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Energy-consumption model for rotary-wing drones

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Abstract

With technological advancement, the use of drones in delivery systems has become increasingly feasible. Many companies have developed rotary-wing drone (RWD) technologies for parcel delivery. At present, the limited endurance is the main disadvantage of RWD delivery. The energy consumption of RWDs must be carefully managed, and it is necessary to develop an effective energy-consumption model to support RWD flight planning. Because the interaction between the forces on the RWD and its flying environment is very complex, it is challenging to estimate accurately the RWD energy consumption. This study summarizes several energy-consumption models proposed in the literature, then we develop an RWD energy-consumption model (called the integrated model) based on analyzing the dynamic equilibrium of forces and power consumption in flight phases (including climb, descent, hover, and horizontal flight). Computational experiments involving several commercial RWDs indicate that the integrated model is more effective than several models in the literature. In the case where an RWD completed one flight segment, on average, 87.63% of the battery capacity was consumed in the horizontal flight phase. We also analyzed the effects of the total mass and horizontal airspeed on the RWD endurance and found that a larger mass corresponded to shorter endurance, and in the experimental range of the horizontal airspeed, a higher horizontal airspeed corresponded to longer endurance. Moreover, the total mass affected the RWD endurance more significantly than the horizontal airspeed.

KEYWORDS

drone energy model, energy consumption, flight phases, rotary-wing drone

1 | INTRODUCTION

Considering the shape of fuselage, several configurations are widely used in the design of drones, for example, fixed-wing, rotary-wing, and hybrid-wing (Macrina et al., 2020). Fixed-wing drones can carry larger payloads for longer distances, but they require constant air movement during flight. The most common rotary-wing drones (RWDs) are the quadcopter, hexacopter, and octocopter. RWDs can be launched and retrieved in a small place. The application fields of drones can be divided into civilian, environmental, military, and so forth. Last-mile parcel delivery is one of the most promising drone applications. Many companies have been developing drone delivery

modes based on RWD technologies (Chung et al., 2020; Macrina et al., 2020; Murray & Chu, 2015). The potential advantages and disadvantages of RWD delivery modes, such as flexible launch/retrieval, high speed compared with ground-vehicle delivery, limited endurance, and limited capacity, have been identified.

For logistics enterprises and customers, delivery costs and time are the most important concerns. Failure to meet punctuality guarantees can lead to penalties and loss of business. In the scenarios of RWDs being adopted for parcel deliveries, the RWD energy consumption is a key variable because it affects the route plans and the quality of delivery services. In an RWD delivery task, the RWD spends time at a selected airspeed. A tradeoff between energy

consumption and performance must be achieved; thus, an effective energy-consumption model that supports considerate route planning for given RWD configurations is necessary.

In the near future, the energy supply on RWDs will remain limited because of many factors, such as the fuselage design and battery technologies. It is important to carefully manage the energy consumption to dispatch RWDs so that they can visit safely the assigned node(s) and to schedule RWDs to deliver as many parcels as possible. An accurate estimation of RWD energy consumption guarantees the feasibility of RWD delivery as well as the efficient operation of RWDs. However, estimating RWD energy consumption is challenging because the interactions between the forces on RWDs and the environment are complex. In the literature, mathematical programming models for optimizing drone or truck-drone routes incorporate energy consumption by applying a predetermined limit on the drone endurance (e.g., Chiang et al., 2019; Kitjacharoenchai et al., 2020; Kloster et al., 2023; Murray & Chu, 2015). In some studies on drone delivery systems, energy-consumption models based on analyzing the physical forces on drones were used (e.g., Dorling et al., 2016; Figliozzi, 2017; Kirschstein, 2020; Kong et al., 2022; Luo et al., 2023; Murray & Raj, 2020; Poikonen & Golden, 2020; Stolaroff et al., 2018; Tran et al., 2020; Wen et al., 2022; Wu et al., 2022).

In this paper, we summarized several energy-consumption models proposed in the literature, and developed an RWD energy-consumption model (named integrated model) from an integrated perspective by analyzing the dynamic equilibrium of forces and power consumption in flight phases. The integrated model consists of the energy consumption in climbing, descending, hovering, and horizontal flight phases. From the methodological perspective, the integrated model encompasses several features of the other methods, making it possible to consider all contributions to energy consumption for each specific flight phase. Computational experiments involving several commercial RWDs indicated that the results of the integrated model developed in this study can be more comprehensive than several models presented in the literature. Logistics companies can adopt the integrated model to estimate the RWD energy consumption, so as to carefully plan RWD routes.

The remainder of this paper is organized as follows. We review the conclusions in the literature from several perspectives in Section 2. The integrated model is developed, and the method for estimating the energy consumption of one flight segment is presented in Section 3. Sections 4 and 5 describe the computational experiments. Section 4 presents parameter values of several commercial RWDs, followed by the experiments for estimating the energy consumption under the conditions of horizontal flight and various phases of one flight segment. Section 5 discusses the effects of the total mass and horizontal airspeed on the RWD endurance. Section 6 presents the conclusions.

For reader convenience, the parameters and variables used in the paper are listed in Table A1.

2 | LITERATURE REVIEW

This section reviews previous studies from several perspectives. It is divided into two subsections pertaining to the factors that affect RWD energy consumption, and energy-consumption models.

2.1 | Factors that affect RWD energy consumption

Several factors affect RWD energy consumption. We classify them into three categories: design-, flying-, and environment-related. Design-related factors include the mass (determined by the battery, drone, and payload), width (including the RWD width and blade chord width), radius (e.g., propeller radius, rotor radius, radius of the whole rotor blade), area (e.g., spinning blade disc area, projected frontal area, propeller disc area), battery, and efficiency (e.g., rotorcraft propulsion system efficiency). Flying-related factors include the speed (e.g., horizontal airspeed, angular speed of the blade, rotor blade tip speed) and angle of attack. Environment-related factors include the air density, wind direction and speed, and temperature.

Complex relationships exist between the RWD energy consumption and these factors, as well as among several factors, as shown in Figure 1. For example, (i) Increasing the RWD mass increases the energy consumption rate (Zhang et al., 2021) and necessitates higher angles of attack. (ii) The energy consumption is affected by the efficiency of power transfer from the battery to the propellers (Figliozzi, 2017). (iii) Larger propellers are efficient. However, they have more inertia and more kinetic energy. Thus, the rotor size is limited (Stolaroff et al., 2018; Zeng et al., 2019). (iv) Both the lift coefficient and the drag coefficient are functions of the angle of attack. Within a certain range, a higher angle of attack corresponds to larger lift and drag coefficients. However, when the angle of attack increases beyond a certain value (called the critical angle of attack), the lift coefficient decreases. (v) The number of rotors affects the propeller disc area. If other conditions remain unchanged, a larger number of rotors corresponds to a larger area. (vi) A higher degree of rotor solidity corresponds to a higher profile power and, thus, more energy consumption (Kirschstein, 2020; Yan et al., 2021; Zeng et al., 2019). (vii) The rotor blade tip speed is equal to the angular speed of blade multiplied by radius of the blade and is proportional to the induced power consumption. (viii) A higher induced velocity corresponds to a higher induced power and more energy consumption (Sorbelli et al., 2020; Stolaroff et al., 2018).

2.1.1 | Mass

The drone (including onboard battery), and payload masses make up of the total mass of the drone and affect the fuselage design and schedule decisions. Increasing the drone mass leads to a higher power requirement (Kyriakakis et al., 2023; Momeni et al., 2023; Rotaru & Todorov, 2018; Zhang et al., 2021). Sorbelli et al. (2020) estimated the energy consumed for covering an arc of per unit distance by a delivery octocopter under different wind conditions,

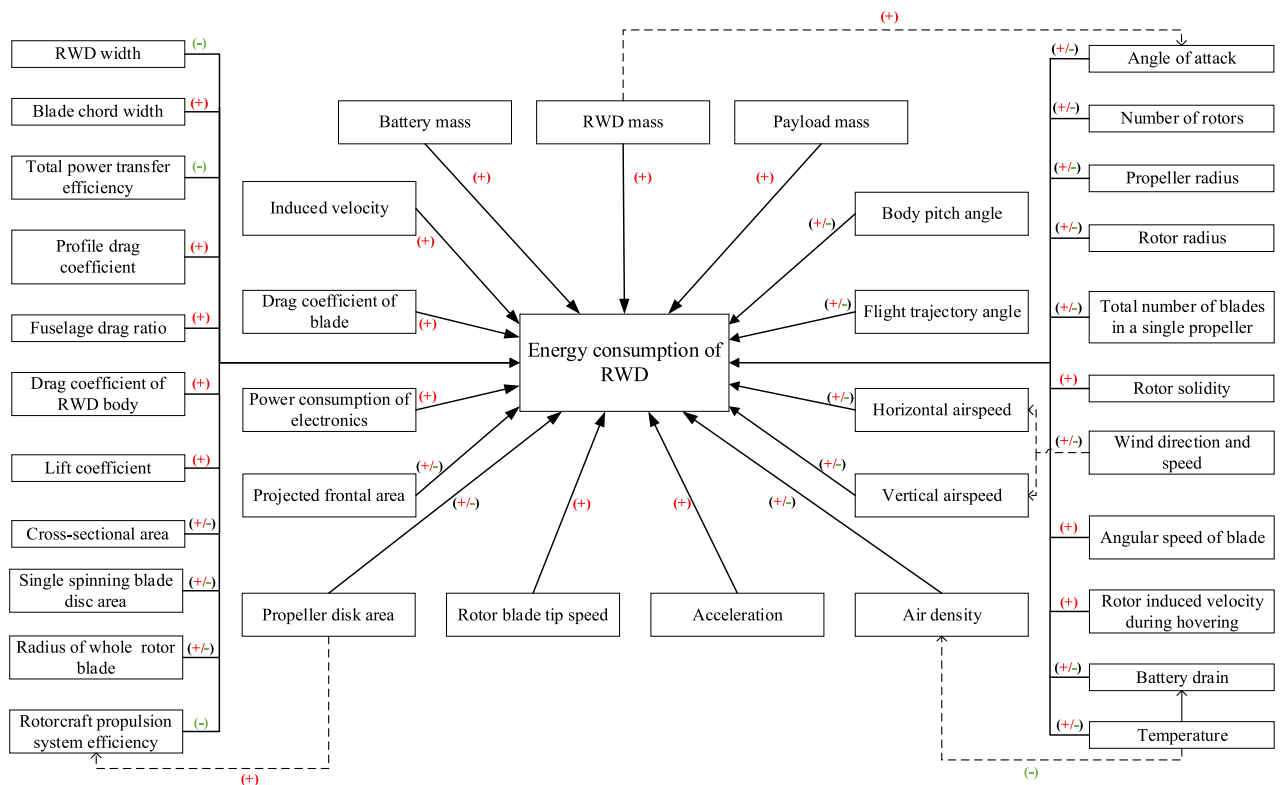


FIGURE 1 Relationships among the rotary-wing drone (RWD) energy consumption and different factors. Solid lines indicate relationships between the RWD energy consumption and various factors. Dashed lines indicate relationships between different factors.

payloads, and flying airspeeds. They observed that a larger mass corresponded to more energy consumption. Choi and Schonfeld (2017) assumed that the energy consumption increased in proportion to the drone and parcel mass. Jeong et al. (2019) assumed that the power consumed increased nearly linearly as the payload increased. Thibbotuwawa et al. (2019a) reported that the highly nonlinear relationship for both airspeed and payload was clear.

2.1.2 | Airspeed

The power consumed is generally a convex function of the airspeed (Zhang et al., 2021). When the RWD vertical takeoff or landing (VTOL) airspeed is about zero, the thrust decreases to gravity. Variations in the horizontal airspeed affect the energy consumption significantly and, consequently, the flight endurance. Considering the drag forces, the power consumed increases at high airspeeds (Chen et al., 2018). At lower airspeeds, the energy consumption has a convex nonlinear relationship with the airspeed. For higher airspeeds, the relationship between the energy consumption and airspeed tends to be linear (Thibbotuwawa et al., 2019a).

In the literature, drone acceleration/deceleration is generally not considered. For example, Chen et al. (2018) ignored the acceleration and deceleration during VTOL and horizontal flight.

To avoid a dynamic model, Liu et al. (2017) assumed that the RWD was in the steady state with zero net acceleration. Considering scenarios where the RWD maneuvering only accounted for a small part of the operating time, Zeng et al. (2019) ignored the energy consumption needed by acceleration/deceleration. Stolaroff et al. (2018) ignored the data on the drone slowing or accelerating.

2.1.3 | Motor and rotor

The power that originates from the propulsive system is delivered to the rotors. However, part of the power is spent for other purposes. The efficiencies of the motor in relation to angular speed and torque should be considered. The power consumption can be accurately estimated by referring to the motor's angular speed. A higher motor angular speed corresponds to a higher thrust (Chen et al., 2018).

Larger propellers are efficient while they are less responsive. Multiple rotors must be spaced so that they do not interfere with each other. These factors limit the rotor size (Stolaroff et al., 2018). Cargo helicopters have large-diameter rotors or multiple rotors, while the smaller rotors employed by quadrotors and multirotors are more energy-intensive (Driessens & Pounds, 2013).

2.1.4 | Environmental factors

Ignoring the effects of environmental factors (e.g., wind and weather) on the energy consumption can lead to unrealistic solutions. However, it is challenging to consider environmental factors in RWD energy-consumption models, because (i) data on environmental factors (e.g., wind data), which are not mature or reliable, are required as one of the model inputs and (ii) the effects of environmental factors vary on small temporal/spatial scales, making it difficult for RWD energy-consumption models to capture them.

Wind direction and speed affect the RWD hovering and climbing. When there is relative airflow over the rotor disc, translational lift appears. Wind can benefit the RWD energy consumption or increase the resistance to RWD flights. Headwind is desirable because it provides the most significant performance enhancement among the different wind directions. Less power consumption is observed when an RWD transitions from hovering to horizontal flight with a headwind. An RWD flying into a headwind leads to an increase in the translational lift, which results in less power consumption. Flying with the wind can reduce the energy consumption in some cases (Dorling et al., 2016; Thibbotuwawa et al., 2019b). Kirschstein (2020) analyzed two wind settings: constant crosswind and head/tailwind. Under the crosswind condition, RWDs must compensate for the air drag to maintain their direction; thus, the energy consumption increased. In the case of a strong and steady crosswind, the RWD endurance is significantly reduced, or the RWD may become unstable (Stolaroff et al., 2018). In the horizontal flight phase, regarding head/tailwind, tailwind supports the RWD, reducing the amount of energy needed, whereas headwind increases the amount of energy needed.

As the altitude increases, the air density decreases. Less power is available at a less density. Increasing the density altitude increases the power required for hovering and flight with low airspeeds. At higher airspeeds, the lower air densities lead to lower power consumed. Thibbotuwawa et al. (2019a) observed that a linear relationship with high air densities increased the energy consumption.

The air density is directly related to the atmospheric temperature. Different atmospheric temperature conditions lead to air density changes, affecting the power consumption of RWDs. The relationship between the temperature and air density is linked with the battery drain. In cold conditions, that is, $\leq 0^\circ\text{C}$, RWD flying times become shorter, and there is an increased risk of RWD malfunction. Temperature can impact the battery performance, because it affects the battery drain and capacity.

2.2 | Energy-consumption models

Several RWD energy-consumption models are presented in the literature, which are classified into four types in this study: simplified models, models based on hovering thrust, models based on horizontal thrust, and models based on static equilibrium.

2.2.1 | Simplified models

1. In several studies, the RWD energy consumption was estimated according to the energy consumption per unit distance. For example, Zorbas et al. (2013) assumed that a larger vertical distance d_v corresponded to more energy consumption and estimated the RWD energy consumption as Equation (1), where E denotes the total energy consumption, m_d denotes the mass of RWD, and g denotes gravitational acceleration.

$$E = m_d g d_v. \quad (1)$$

Ahmed et al. (2016) assumed that RWDs moved horizontally across a segment, during which they consumed E_s^d per unit horizontal distance, and climbed or descended vertically, during which they consumed E_c^d or E_d^d , respectively, per unit vertical distance. The total energy consumption was given as Equation (2), where d_s denotes the horizontal flight distance, E_h^t denotes the energy consumption per unit hovering time, and t_h is the hovering time.

$$E = E_s^d d_s + (E_c^d + E_d^d) d_v + E_h^t t_h. \quad (2)$$

Rabta et al. (2018) and Kyriakakis et al. (2023) considered that energy consumption is related to both payload and distance. Rabta et al. (2018) proposed Equation (3), where m_p denotes the mass of payload, E_v and E_s denote the energy consumption in vertical and horizontal flight, respectively, k_1 represented the energy consumed in course of takeoff/landing with package(s), k_2 represented the energy consumed in course of takeoff/landing for the empty RWD, k_3 represented the energy consumed for flying one distance unit with package(s), and k_4 represented the energy consumed for flying one distance unit for the empty RWD.

$$E(m_p) = E_v + E_s = (k_1 m_p + k_2) + (k_3 m_p + k_4) d_s. \quad (3)$$

2. In several studies (e.g., Choi & Schonfeld, 2017; D'Andrea, 2014; Figliozzi, 2017), the RWD energy consumption was estimated according to the lift-to-drag ratio. Maintaining steady horizontal flight requires a force equilibrium of an RWD. There are generally four forces acting on an RWD: the gravitational force (G), that is, the gravitational force acting downward; thrust (T), that is, the propelling force; lift (L), that is, the force acting at a right angle to the motion direction; and drag (D), that is, the force acting opposite to the motion direction. Lift and thrust are integrated in the propellers. The L/D ratio, also named the lift-to-drag ratio (denoted as r_{ld}), is a key factor that can affect the flight efficiency. The energy needed for horizontal flight is given by Equation (4), where T_s denotes the thrust during horizontal flight, v_s denotes the horizontal airspeed, t_s denotes the horizontal flight time, m denotes the total mass ($m = m_b + m_p + m_d$, and m_b is the mass of battery), and e_t denotes the power transfer efficiency. In Equation (4), the L/D ratio is typically a function of the airspeed.

$$E_s = T_s v_s t_s = mg d_s / (r_{ld} e_t). \quad (4)$$

2.2.2 | Models based on hovering thrust

In the literature, models based on hovering thrust (e.g., Choi et al., 2020; Murray & Raj, 2020; Thibbotuwawa et al., 2019b; Xia et al., 2023) assume that the power consumed in horizontal flight is reduced owing to translational lift, which is the conversion from the hover to forward flight. Therefore, the power for hovering can be regarded as an upper bound of the power for horizontal flight. The power for takeoff/landing is assumed to equal the power for hovering. Because the thrust to oppose G is provided by the induced air passing the propellers, according to the actuator disc theory, the thrust (denoted as T_h) is given by Equation (5), where ρ denotes the air density, A_p denotes the propeller disc area, v_{ih} denotes the induced velocity during hovering.

$$T_h = 2\rho A_p v_{ih}^2. \quad (5)$$

The corresponding power (denoted as P) is given by Equation (6). The energy consumption is determined by the power consumption and total flight time t , that is, $E = P \cdot t$.

$$P = T_h \cdot v_{ih} = \sqrt{T_h^3 / (2\rho A_p)}. \quad (6)$$

Trihinas et al. (2021) considered especially the motor efficiency and adopted Equation (7).

$$P = \sqrt{T_h^3 / (2\rho A_p)} / e_t. \quad (7)$$

Dorling et al. (2016) and Xia et al. (2023) assumed $T_h = mg$. Under the configuration of a certain hexacopter, the multi-copter power consumption is calculated as Equation (8), where $k_5 = 46.7$ W/kg and $k_6 = 26.9$ W.

$$P = k_5 m_p + k_6. \quad (8)$$

Referring to Dorling et al. (2016), Murray and Raj (2020), and Mahmoudi and Eshghi (2022) assumed that the power consumed during climbing, horizontal flight, descending, and hovering were identical (Note: This assumption is not true in actual situations) and could be calculated as Equation (9), where $k_7 = 210.8$ W/kg and $k_8 = 181.2$ W.

$$P = k_7 m_p + k_8. \quad (9)$$

Referring to Dorling et al. (2016), Jeong et al. (2019) considered the energy consumption and ignored the takeoff/landing course. Cheng et al. (2020) used the power consumption during hovering, which was set via a linear approximation, to approximate the horizontal power consumption.

2.2.3 | Models based on horizontal thrust

Models based on horizontal thrust assume that the energy consumption equals to the work done by the thrust during the horizontal flight, and the thrust is derived from the force equilibrium. Based on the actuator disc theory, the power is calculated using Equation (10), and the induced velocity during horizontal flight (denoted as v_{is}) can be determined by solving the implicit Equation (11), where T_s denotes the thrust during horizontal flight, v is a given airspeed, and α represents the tilt angle of the rotor disc.

$$P = T_s (v_{is} + v \cdot \sin \alpha), \quad (10)$$

$$v_{is} = T_s / (2\rho A_p \sqrt{(v \cdot \cos \alpha)^2 + (v \cdot \sin \alpha + v_{is})^2}). \quad (11)$$

In forward horizontal flight, the RWD is slightly inclined, and the tilt angle of the rotor disc is calculated as Equation (12), where D_s denotes the drag force in horizontal flight.

$$\alpha = \tan^{-1}(D_s / G). \quad (12)$$

In several studies (e.g., Sorbelli et al., 2020; Stolaroff et al., 2018; Tamke & Buscher, 2023), the thrust was estimated using Equation (13), and the tilt angle of the rotor disc is calculated as Equation (14), where C_d denotes the drag coefficient of RWD body, A_f denotes the projected frontal area, and v_s denotes the horizontal airspeed. Equation (10) is corrected by e_t , as indicated by Equation (15).

$$T_s = G + D_s = mg + \rho C_d A_f v_s^2 / 2, \quad (13)$$

$$\alpha = \tan^{-1}((\rho C_d A_f v_s^2 / 2) / (mg)), \quad (14)$$

$$P = T_s (v_{is} + v \cdot \sin \alpha) / e_t. \quad (15)$$

2.2.4 | Models based on static equilibrium

Models based on static equilibrium assume that the RWD is in a steady state and that the external forces are in static equilibrium, yielding zero net acceleration. Power is consumed by the RWD during horizontal flight, takeoff, and landing. The power is classified into three types: induced power (P_{in}), profile power (P_{pr}), and parasite power (P_{pa}). The induced power generates thrust by propelling air downward, the profile power overcomes the rotational drag encountered by the rotating propeller blades, and the parasite power resists body drag when there is a relative translational motion between the RWD and wind. In some studies, the power consumption of electronic components was considered.

Thibbotuwawa et al. (2019b) analyzed horizontal flight with a constant airspeed. In this case, all the forces balance each other so that the net force is zero; that is, the upward lift equals the downward force of gravity, and the forward thrust of the rotors is

balanced by the backward drag. The induced power is estimated as Equation (16), which is related to the total mass, air density, RWD width (denoted as w_d), and horizontal airspeed. The parasite power is estimated as Equation (17). The profile power is ignored. So, the total power consumption during horizontal flight is estimated as Equation (18).

$$P_{in} = (mg)^2 / (\rho v_s w_d^2), \quad (16)$$

$$P_{pa} = \rho C_d A_f v_s^3 / 2, \quad (17)$$

$$P = P_{in} + P_{pa}. \quad (18)$$

Considering that all the forces must be balanced when the RWD is in horizontal and steady-level flight, Hwang et al. (2018) estimated the thrust as Equation (19). The induced power is estimated as Equation (20), where v_{is} is estimated using Equation (11). The parasite power is estimated as Equation (21).

$$T_s = \sqrt{G^2 + D_s^2} = \sqrt{(mg)^2 + (\rho C_d A_f v_s^2 / 2)^2}, \quad (19)$$

$$P_{in} = T_s v_{is}, \quad (20)$$

$$P_{pa} = D_s v_s. \quad (21)$$

Ignoring the RWD acceleration/deceleration, Zeng et al. (2019) stated that the blade profile power and parasite power increased quadratically and cubically, respectively, with an increase in v_s . The induced power decreases with an increase in v_s . For an RWD flying forward with airspeed v_s , the power consumption is estimated as Equation (22), where k_9 is the incremental correction factor to correct the deviation between the theoretical and practical induced power, C_b denotes the drag coefficient of blade, s_r denotes the rotor solidity, ω_b denotes the angular speed of blade, R_r denotes the radius of signal rotor, v_r denotes the rotor blade tip speed, and r_d is the fuselage drag ratio.

$$P(v_s) = (1 + k_9) m_d g (\sqrt{1 + v_s^4 / (4v_{ih}^4)} - v_s^2 / (2v_{ih}^2))^{1/2} / (2\rho A_p)^{1/2} + \rho C_b s_r A_p \omega_b^3 R_r^3 (1 + 3v_s^2 / v_r^2) / 8 + \rho r_d s_r A_p v_s^3 / 2. \quad (22)$$

We set $v_s = 0$; then, the power consumption for hovering (P_h) is estimated as Equation (23).

$$P_h = (1 + k_9) m_d g / (2\rho A_p)^{1/2} + \rho C_b s_r A_p \omega_b^3 R_r^3 / 8. \quad (23)$$

As v_s increases, $P(v_s)$ first decreases and then increases. When $v_s \gg v_{ih}$, according to the first-order Taylor approximation, the induced power consumption during forward flight is changed as Equation (24).

$$P_{in} = (1 + k_9) m_d g v_{ih} / (v_s (2\rho A_p)^{1/2}). \quad (24)$$

Yan et al. (2021) adjusted the model developed by Zeng et al. (2019) by considering the acceleration/deceleration with the initial

airspeed. The calculation methods for the induced power and profile power remain approximately the same, except that the airspeed is changed from v_s to $v_0 + a_s t$, where v_0 denotes the initial airspeed, and a_s denotes the acceleration during forward horizontal flight. The parasite power is estimated as Equation (25) (Yan et al., 2021), where e_o is the Oswald efficiency with typical value between 0.70 and 0.85, and r_w denotes the aspect ratio of wing.

$$P_{pa} = \rho C_d A_f v_s^3 / 2 + 2(m_d g)^2 / (\pi \rho e_o r_w A_f v_s) + m_d a_s v_s. \quad (25)$$

Liu et al. (2017) developed a power model consisting of P_{in} , P_{pr} , and P_{pa} . Campuzano et al. (2023) adopted the model presented in Liu et al. (2017) to estimate the energy consumption. P_{in} and P_{pr} are calculated through Equations (26) and (27), respectively, where k_{10} is a scaling factor, $0 < k_{10} < 1$; k_{11} is a scaling factor, T denotes the thrust in the steady state, v_v denotes the vertical airspeed, N_b denotes the total number of blades in single propeller, and w_b denotes the blade chord width.

$$P_{in} = k_{10} T (v_v / 2 + \sqrt{(v_v / 2)^2 + T / (2\rho A_p)}), \quad (26)$$

$$P_{pr} = k_{11} \rho C_b N_b w_b R_r^4 T^{3/2} / 8. \quad (27)$$

The thrust in the steady state (T) is estimated as Equation (28), where C_{in} denotes the coefficient of induced drag.

$$T = \sqrt{(m_d g - \rho C_{in} N_b w_b R_r (v_s \cos \alpha)^2)^2 / 4 + (\rho C_d A_f v_s^2)^2} / 4. \quad (28)$$

During vertical flight, $T = m_d g$, and the power consumption is $P = P_{in} + P_{pr}$. During horizontal flight, $v_v = 0$, and the power consumption is $P = P_{in} + P_{pr} + P_{pa}$. During hovering, $T = m_d g$ and $v_v = 0$, and the power consumption is $P = P_{in} + P_{pr}$.

Kirschstein (2020) considered four phases of RWD flight: takeoff, horizontal flight, hovering, and landing. The power required for overcoming the RWD-body's and rotors' air drag (i.e., P_{pa} and P_{pr} , respectively) and for lifting (P_{lift}), climbing (P_c), and internal electronics (P_e) are calculated as Equation (29), where κ is an upscaling factor to correct the deviation between the idealized uniform airflow and actual airflow, C_w denotes the downwash coefficient, and θ_c denotes the flight trajectory angle in climbing. The thrust is calculated as Equation (30).

$$P = P_{pa} + P_{pr} + P_{lift} + P_c + P_e = \rho C_d A_f v^3 / 2 + \rho C_b s_r A_p v_r^3 (1 + 2v^2 / v_r^2) / 8 + \kappa C_w T + mgv \sin \theta_c + P_e, \quad (29)$$

$$T = \sqrt{(mg)^2 + (\rho C_d A_f v^2 / 2)^2} + mg \rho C_d A_f v^3 \sin \theta_c. \quad (30)$$

The power (P_c) consumed during VTOL (i.e., $\theta_c = 90^\circ$) is estimated as Equation (31), where v_c denotes the climbing airspeed. The power (P_h) consumed during hovering (i.e., $v_c = 0$) is estimated as Equation (32), and the power (P_s) consumed during horizontal flight (i.e., $\theta_c = 0^\circ$) is estimated as Equation (33).

$$P_c = \rho C_d A_f v_c^3 / 2 + \rho C_b s_r A_p v_r^3 (1 + 2v_c^2 / v_r^2) / 8 + \kappa C_w \sqrt{(mg)^2 + (\rho C_d A_f v_c^2 / 2)^2} + mg \rho C_d A_f v_c^3 + 0.1, \quad (31)$$

$$P_h = \rho C_b s_r A_p v_r^3 / 8 + \kappa C_w mg + 0.1, \quad (32)$$

$$P_s = \rho C_d A_f v_s^3 / 2 + \rho C_b s_r A_p v_r^3 (1 + 2v_s^2 / v_r^2) / 8 + \kappa C_w \sqrt{(mg)^2 + (\rho C_d A_f v_s^2 / 2)^2} + 0.1. \quad (33)$$

streamtube (i.e., an axially symmetric surface), (ii) incompressible air and a constant airflow, and (iii) the airflow remaining in the same direction. Figure 2 shows the streamtube cross-sectional areas in various phases, where vertical takeoff and vertical descending are special cases of climbing and descending, respectively. Figure 3 presents an analysis of the dynamic equilibrium of forces acting on the RWD. The power consumption can be estimated according to the conservation of mass, momentum, and energy.

3 | INTEGRATED MODEL BASED ON DYNAMIC EQUILIBRIUM

Considering the RWD flying conditions, we refer to actuator disc theory to analyze the dynamic course of RWD flight phases. The integrated model is derived according to the streamtube cross-sectional areas in flight phases and the force analysis.

3.1 | Dynamic equilibrium of forces

According to actuator disc theory, a lifting force is achieved by a change in momentum. The conditions required include (i) the existence of a

3.2 | Power consumption

1. In the phase of RWD hovering (Figure 2a), the airflow enters the streamtube; then, it is accelerated by the rotor disc; finally, it is exhausted from the bottom of the streamtube. Far upstream of the disc, the vertical velocity (denoted as v_1) of the airflow tends to zero, that is, $v_1 = 0$. v_2 denotes the velocity of airflow exhausted from bottom of streamtube. The thrust, that is, T_h , is calculated based on the increase in momentum as shown in Equation (34), where $\dot{m}$ represents the mass of the airflow per unit time and is calculated as Equation (35).

$$T_h = \dot{m}(v_2 - v_1) = \dot{m} v_2 = \rho A_p v_{ih} \cdot v_2, \quad (34)$$

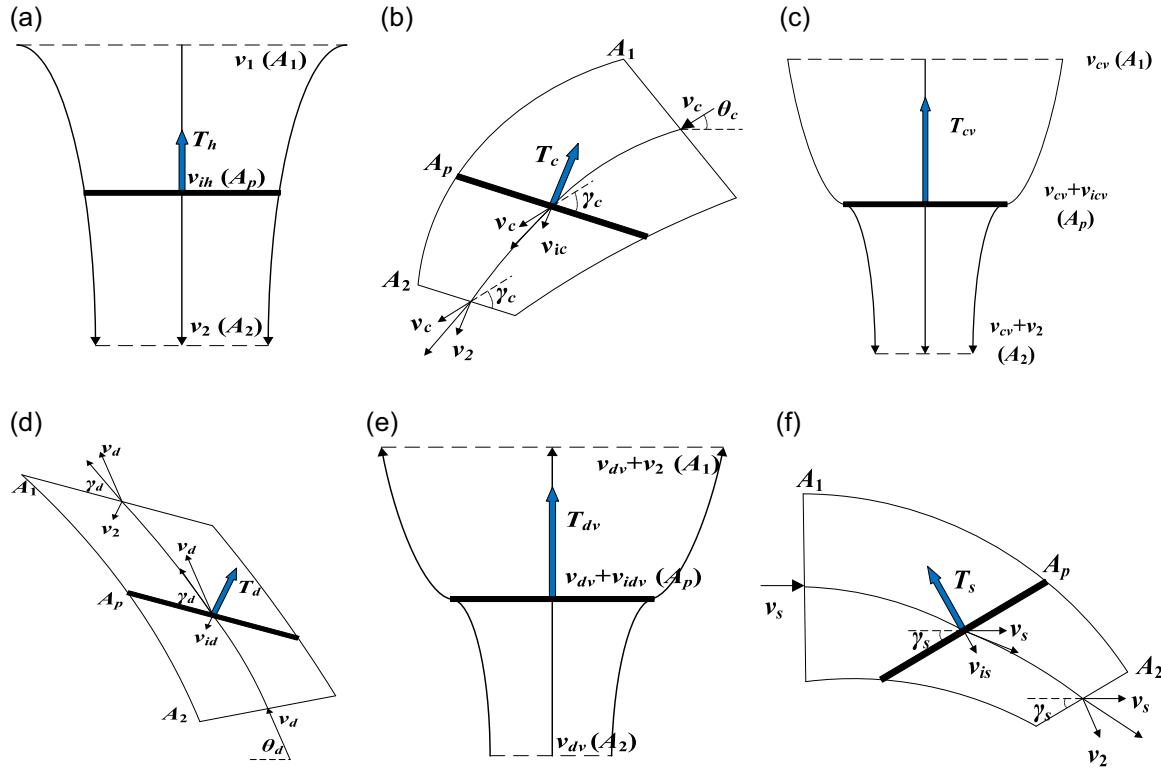


FIGURE 2 Streamtube cross-sectional areas in flight phases. (a) Hovering, (b) climbing ($\theta_c \neq 90^\circ$), (c) vertical takeoff ($\theta_c = 90^\circ$), (d) descending ($\theta_c \neq 90^\circ$), (e) vertical descending ($\theta_c = 90^\circ$), and (f) horizontal forward. A_1 denotes the cross-sectional area where airflow enters streamtube; A_2 denotes the cross-sectional area where airflow is exhausted from bottom of streamtube; v_{cv} denotes the vertical-takeoff airspeed; v_{icv} denotes the induced velocity during vertical takeoff; T_{cv} denotes the thrust during vertical takeoff; v_{dv} denotes the descending vertical-descending airspeed; v_{idv} denotes the induced velocity during vertical descending; T_{dv} denotes the thrust during vertical descending.

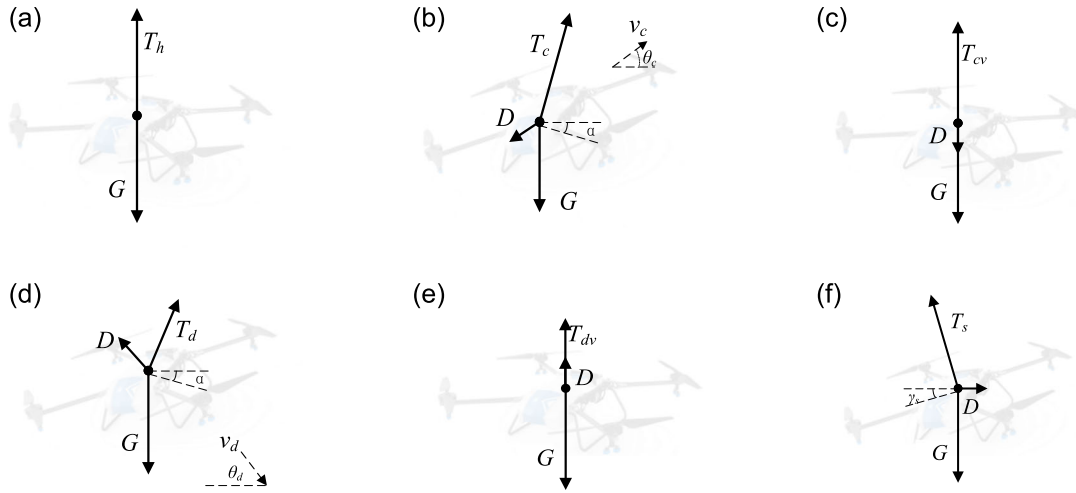


FIGURE 3 Force analysis for flight phases. (a) Hovering, (b) climbing, (c) vertical takeoff, (d) descending, (e) vertical descending, and (f) horizontal forward.

$$\dot{m} = \rho A_1 v_1 = \rho A_p v_{ih} = \rho A_2 v_2. \quad (35)$$

The thrust can also be estimated using the difference in air pressure between the two sides of the rotor disc. Considering the energy balance given by Equation (36), the induced velocity is doubled when the airflow forms the wake, that is, $v_2 = 2v_{ih}$.

$$P = \dot{m} v_2^2 / 2. \quad (36)$$

The thrust is given by Equation (37), and v_{ih} is calculated as Equation (38).

$$T_h = 2\rho A_p v_{ih}^2, \quad (37)$$

$$v_{ih} = \sqrt{T_h / (2\rho A_p)}. \quad (38)$$

The induced power is estimated as Equation (39), and the profile power is estimated as Equation (40).

$$P_{in} = T_h v_{ih} = T_h \cdot \sqrt{T_h / (2\rho A_p)}, \quad (39)$$

$$P_{pr} = \rho C_b A_p v_r^3 / 8. \quad (40)$$

The method for calculating the profile power is referred to Zeng et al. (2019), and Kirschstein (2020). Then, the total power is estimated as Equation (41).

$$P_h = T_h \cdot \sqrt{T_h / (2\rho A_p)} + \rho C_b A_p v_r^3 / 8 + P_e. \quad (41)$$

Under the conditions of hovering and zero wind speed (Figure 3(a)), we have Equations (42) and (43).

$$T_h = G = mg, \quad (42)$$

2. RWD climbing requires a momentum change and an upward thrust perpendicular to the rotor (Figure 2b). The airflow enters the streamtube far upstream of the rotor; then, it passes through the rotor; finally, it moves away from the rotor and forms the wake. Assume that the airflow enters the streamtube with velocity v_c . Then, the airflow undergoes a velocity increase of v_{ic} when it passes through the rotor disc. The airflow finally forms the wake with velocity v_c to v_2 . The thrust is given by Equation (44), and $\dot{m}$, γ_c and v_2 are calculated as Equations (45)–(47), where a_c denotes the acceleration during climbing, t_c is the climbing time, γ_c denotes the body pitch angle during climbing, and v_{ic} denotes the induced velocity during climbing.

$$T_c = \dot{m} \vec{v}_2 = \rho A_p \sqrt{((v_c + a_c t_c) \cos \gamma_c)^2 + ((v_c + a_c t_c) \sin \gamma_c + v_{ic})^2} \cdot v_2, \quad (44)$$

$$\dot{m} = \rho A_1 (\vec{v}_c + \vec{a}_c t_c) = \rho A_p ((\vec{v}_c + \vec{a}_c t_c) + \vec{v}_{ic}) = \rho A_2 ((\vec{v}_c + \vec{a}_c t_c) + \vec{v}_2), \quad (45)$$

$$\gamma_c = \theta_c + \alpha, \quad (46)$$

$$v_2 = 2v_{ic}. \quad (47)$$

The induced power, the profile power and the parasite power are given by Equations (48)–(50).

$$P_{in} = T_c \cdot ((v_c + a_c t_c) \sin \gamma_c + v_{ic}), \quad (48)$$

$$P_{pr} = \rho C_b A_p v_r^3 (1 + 3(v_c + a_c t_c)^2 / v_r^2) / 8, \quad (49)$$

$$P_{pa} = \rho A_f (v_c + a_c t_c)^3 / 2. \quad (50)$$

The method for calculating the parasite power is referred to Liu et al. (2017) and Kirschstein (2020). Then, the total power is estimated as Equation (51).

$$P = T_c \cdot ((v_c + a_c t_c) \sin \gamma_c + v_{ic}) + \rho C_b A_p v_r^3 (1 + 3(v_c + a_c t_c)^2 / v_r^2) / 8 + \rho A_f (v_c + a_c t_c)^3 / 2 + P_e. \quad (51)$$

Under the conditions of climbing and the equilibrium of forces (Figure 3b), T_c can be calculated as Equation (52).

$$T_c = \begin{cases} \sqrt{(mg \cdot \sin \theta_c + \rho A_f v_c^2 / 2)^2 + (mg \cdot \cos \theta_c)^2}, & \text{if } a_c = 0 \\ \sqrt{(ma_c + mg \cdot \sin \theta_c + \rho A_f (v_c + a_c t_c)^2 / 2)^2 + (mg \cdot \cos \theta_c)^2}, & \text{if } a_c \neq 0. \end{cases} \quad (52)$$

- For RWD hovering or climbing, the momentum theory can be used to estimate the rotor thrust, induced velocity, and induced power. However, for RWD descending, the airflow may have large degrees of recirculation and turbulence, making the momentum theory unreliable, that is, when descending airspeed v_d satisfies $-2v_{ih} < v_d < 0$, there is no valid wake model. However, the momentum theory for climbing is valid under the condition of descending at a low airspeed, because there is marginal change in airflow near the rotor and the airflow near the rotor disc is downward. When the descending airspeed is sufficient so that the windmill brake state appears with $-2v_{ih} > v_d$, the airflow becomes smooth.

In the phase of RWD descending (Figure 2d), the airflow below the rotor enters the streamtube with velocity v_d ; then, its velocity is reduced by v_{id} when it passes through the rotor disc. The thrust is given by Equation (53), and $\dot{m}$, γ_d and v_2 are calculated as Equations (54)–(56), where γ_d denotes the body pitch angle during descending, v_{id} denotes the induced velocity during descending, a_d denotes the acceleration during descending, t_d is the time during descending, and θ_d denotes the flight trajectory angle in descending direction.

$$T_d = -\dot{m} \vec{v}_2 = -\rho A_p \sqrt{(v_d \cos \gamma_d)^2 + (v_d \sin \gamma_d - v_{id})^2} \cdot v_2, \quad (53)$$

$$\dot{m} = \rho A_2 (\vec{v}_d + \vec{a}_d t_d) = \rho A_p ((\vec{v}_d + \vec{a}_d t_d) + \vec{v}_{id}) = \rho A_1 ((\vec{v}_d + \vec{a}_d t_d) + \vec{v}_2), \quad (54)$$

$$\gamma_d = \theta_d - \alpha, \quad (55)$$

$$v_2 = 2v_{id}. \quad (56)$$

The induced power, the profile power and the parasite power are given by Equations (57)–(59).

$$P_{in} = T_d \cdot ((v_d + a_d t_d) \sin \gamma_d + v_{id}), \quad (57)$$

$$P_{pr} = \rho C_b A_p v_r^3 (1 + 3(v_d + a_d t_d)^2 / v_r^2) / 8, \quad (58)$$

$$P_{pa} = \rho A_f (v_d + a_d t_d)^3 / 2. \quad (59)$$

Therefore, the total power is estimated as Equation (60).

$$P = T_d \cdot ((v_d + a_d t_d) \sin \gamma_d + v_{id}) + \rho C_b A_p v_r^3 (1 + 3(v_d + a_d t_d)^2 / v_r^2) / 8 + \rho A_f (v_d + a_d t_d)^3 / 2 + P_e. \quad (60)$$

Under the conditions of descending and the equilibrium of forces (Figure 3d), T_d can be calculated as Equation (61).

$$T_d = \begin{cases} \sqrt{(mg \cdot \sin \theta_d - \rho A_f v_d^2 / 2)^2 + (mg \cdot \cos \theta_d)^2}, & \text{if } a_d = 0 \\ \sqrt{(mg \cdot \sin \theta_d - \rho A_f (v_d + a_d t_d)^2 / 2 + ma_d)^2 + (mg \cdot \cos \theta_d)^2}, & \text{if } a_d \neq 0. \end{cases} \quad (61)$$

In this study, all the sample commercial RWDs obey $-\frac{1}{2}v_{ih} < v_d + a_d t_d < 0$.

- RWD forward flight in a certain horizontal plane at a constant airspeed requires a constant thrust for the force equilibrium, so that the streamtube leans forward (Figure 2f). The thrust is given by Equation (62), and $\dot{m}$ and v_2 are calculated as Equations (63) and (64), where γ_s denotes the body pitch angle during horizontal flight.

$$T_s = \dot{m} \vec{v}_2 = \rho A_p \sqrt{((v_s + a_s t_s) \cos \gamma_s)^2 + ((v_s + a_s t_s) \sin \gamma_s + v_{is})^2} \cdot v_2, \quad (62)$$

$$\begin{aligned} \dot{m} &= \rho A_1 (\vec{v}_s + \vec{a}_s t_s) = \rho A_p ((\vec{v}_s + \vec{a}_s t_s) + \vec{v}_{is}) \\ &= \rho A_2 ((\vec{v}_s + \vec{a}_s t_s) + \vec{v}_2), \end{aligned} \quad (63)$$

$$v_2 = 2v_{is}. \quad (64)$$

The dip angle of the rotor disc is generally small; thus, $(v_s + a_s t_s) \sin \gamma_s$ is significantly smaller than v_{is} , and we have Equation (65).

$$(v_s + a_s t_s) \cos \gamma_s \approx v_s + a_s t_s. \quad (65)$$

Hence, v_{is} can be calculated as Equation (66).

$$v_{is} = T_s / (2\rho A_p \sqrt{(v_s + a_s t_s)^2 + v_{is}^2}). \quad (66)$$

The induced power, the profile power and the parasite power are given by Equations (67)–(69).

$$P_{in} = T_s \cdot ((v_s + a_s t_s) \sin \gamma_s + v_{is}), \quad (67)$$

$$P_{pr} = \rho C_b A_p v_r^3 (1 + 3(v_s + a_s t_s)^2 / v_r^2) / 8, \quad (68)$$

$$P_{pa} = \rho A_f (v_s + a_s t_s)^3 / 2. \quad (69)$$

Therefore, the total power is estimated as Equation (70).

$$P = T_s \cdot ((v_s + a_s t_s) \sin \gamma_s + v_{is}) + \rho C_b A_p v_r^3 (1 + 3(v_s + a_s t_s)^2 / v_r^2) / 8 + \rho A_f (v_s + a_s t_s)^3 / 2 + P_e. \quad (70)$$

Under the conditions of horizontal forward flight and the equilibrium of forces (Figure 3f), T_s can be calculated as Equation (71).

$$T_s = \begin{cases} \sqrt{(mg)^2 + (\rho A_f v_s^2 / 2)^2}, & \text{if } a_s = 0 \\ \sqrt{(mg)^2 + (\rho A_f (v_s + a_s t_s)^2 / 2 + m a_s)^2}, & \text{if } a_s \neq 0. \end{cases} \quad (71)$$

Several parts of the power consumption in various phases are presented in Table 1, where the coefficient of profile and parasite drag (denoted as C_{pr} and C_{pa} , respectively) are given as Equations (72) and (73).

$$C_{pr} = \rho C_b A_p v_r^3 / 8, \quad (72)$$

$$C_{pa} = \rho A_f / 2. \quad (73)$$

According to several studies (e.g., Kirschstein, 2020; Langelaan et al., 2017; Liu et al., 2017), the actual induced power loss is 5%–20% higher than the estimate based on momentum theory; thus, an empirical correction factor (denoted as κ , usually equal to 1.15) should be included in the induced-power estimation.

3.3 | Energy consumption of one flight segment

A flight segment begins when an RWD takes off from the ground and ends when the RWD lands on the ground after the flight task. A flight segment consists of climbing, hovering, horizontal flight, and descending. Figure 4 shows the change in the RWD height during one flight segment.

Let the acceleration/deceleration be nonzero. In the course of climbing, from time 0 to time t_1 , the RWD climbs with increasing airspeed and flight trajectory angle θ_c to reach height h_1 , and the final airspeed is v_c . Then, the RWD continues to climb with a constant airspeed v_c to reach height h_2 from t_1 to t_2 . Between t_2 and t_3 , the RWD climbs with a decreasing airspeed to reach height h_3 , and the final climbing airspeed is 0. After the climbing phase, that is, from t_3 to t_4 , the RWD hovers.

In the course of horizontal flight, the flight height remains at h_3 . From t_4 to t_5 , the RWD moves forward with increasing airspeed, and the final airspeed is v_s . Then, the RWD continues to move forward with a constant airspeed, v_s , from t_5 to t_6 . Between t_6 and t_7 , the RWD moves forward with a decreasing airspeed, and the final airspeed is 0. After the horizontal flight phase, that is, from t_7 to t_8 , the RWD hovers.

In the course of descending, the RWD starts at an airspeed of 0 from the hovering point. From t_8 to t_9 , the RWD descends with increasing airspeed and flight trajectory angle θ_d to reach height h_2 , and the final airspeed is v_d . Then, the RWD continues to descend with a constant airspeed v_d to reach height h_1 from t_9 to t_{10} . From t_{10} to t , the RWD lands on the ground with a decreasing airspeed, and the final airspeed is 0.

The energy consumed during RWD climbing (E_c), horizontal flight (E_s), hovering (E_h), and descending (E_d) are estimated, and the energy consumption of one flight segment is estimated as Equation (74).

$$\begin{aligned} E &= E_c + E_s + E_h + E_d \\ &= \int P_c dt_c + \int P_s dt_s + \int P_h dt_h + \int P_d dt_d \\ &= \sum_{j=c,d,s} \int (T_j \cdot (v_j + a_j t_j) \sin \gamma_j + \kappa T_j \cdot v_{ij}) dt_j \\ &\quad + \kappa \sqrt{T_h^3 / (2 \rho A_p)} \cdot t_h \\ &\quad + C_{pr} \cdot (t + 3 \cdot \sum_{j=c,d,s} \int (v_j + a_j t_j)^2 dt_j) / v_r^2 \\ &\quad + C_{pa} \cdot \sum_{j=c,d,s} \int (v_j + a_j t_j)^3 dt_j + P_e \cdot t. \end{aligned} \quad (74)$$

Because of the complexity of Equation (74), we cannot directly draw conclusions on the variation of the energy consumption with respect to different factors except for the total mass. Let m_1 and m_2 denote the total masses, respectively. When $m_1 > m_2$, $E(m_1) > E(m_2)$, and this conclusion is certain.

4 | EFFECTIVENESS OF ENERGY-CONSUMPTION MODELS

4.1 | Parameters of several commercial RWDs

In Section 2, we presented several energy-consumption models from the literature, and in Section 3, we proposed an integrated model. To conduct experiments on the energy-consumption models, we took several commercial RWDs as examples: 3DR H520-G, Parrot ANAFI Ai, DJI Mavic Pro, DJI Mavic 2 Pro, DJI Mavic 3, DJI Mair2, DJI

TABLE 1 Estimation of rotary-wing drone power consumption.

	P_{in}	P_{pr}	P_{pa}
Climbing	$T_c \cdot ((v_c + a_c t_c) \sin \gamma_c + \kappa v_{ic})$	$C_{pr} \cdot (1 + 3(v_c + a_c t_c)^2 / v_r^2)$	$C_{pa} \cdot (v_c + a_c t_c)^3$
Descending	$T_d \cdot ((v_d + a_d t_d) \sin \gamma_d + \kappa v_{id})$	$C_{pr} \cdot (1 + 3(v_d + a_d t_d)^2 / v_r^2)$	$C_{pa} \cdot (v_d + a_d t_d)^3$
Horizontal flight	$T_s \cdot ((v_s + a_s t_s) \sin \gamma_s + \kappa v_{is})$	$C_{pr} \cdot (1 + 3(v_s + a_s t_s)^2 / v_r^2)$	$C_{pa} \cdot (v_s + a_s t_s)^3$
Hovering	$\kappa \sqrt{T_h^3 / (2 \rho A_p)}$	C_{pr}	0

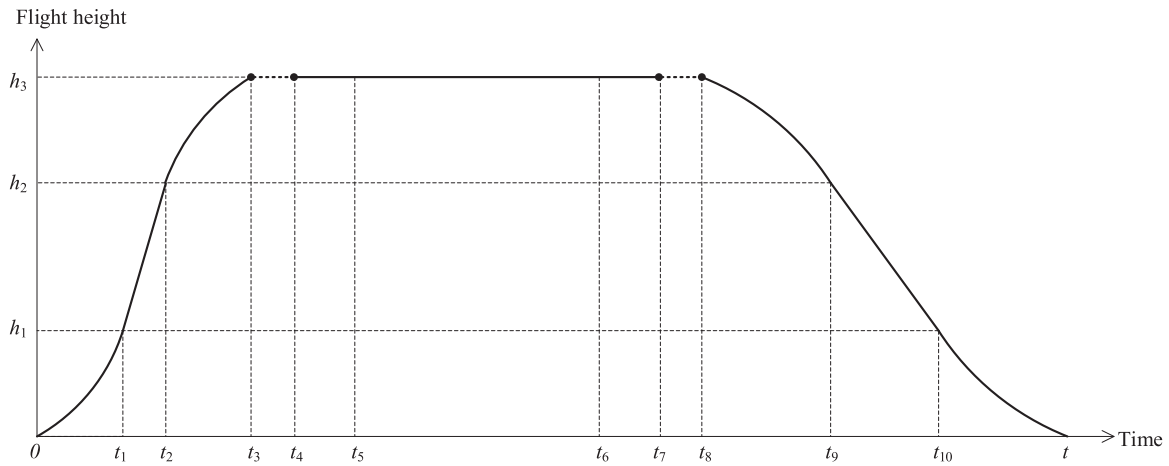


FIGURE 4 Rotary-wing drone height with respect to time.

Phantom 3, DJI Phantom 4, Falcon B, and GD2.0X, which were denoted as RWD1–RWD10, respectively. The example RWDs are generally equipped with Lithium-polymer (Li-Po) batteries, which have a high energy-to-weight ratio, the capability of high-rate discharge, and a variety of shapes and sizes (Avanzini et al., 2016). Considering that the propulsion system has losses in the power output, we introduce power transfer efficiency e_t .

The key parameters of various RWDs were obtained from the manufacturers' official websites, as shown in Table A2. For example, the maximum flight time of RWD5 (DJI Mavic 3) is 2760 s. Specific test conditions are as follows: no wind, at sea level, the constant flight speed of 9 m/s, and flying from 100% battery to 0% battery, and so on. According to the official statements, the values of the RWD parameters were measured in a laboratory environment, that is, the RWD flew at a constant airspeed, v_s , until the battery energy ($E_{battery}$) was exhausted, and the maximum flight time t_{max} was recorded. For RWD2, RWD9, and RWD10, the values of v_r (i.e., $\omega_b \times R_r$) were obtained from the official website, and for the other RWDs, they were set as 120 m/s in accordance with the literature.

In addition to the parameter values provided by the manufacturer, several parameter values were estimated by referring to the literature (Kirschstein, 2020; Langelaan et al., 2017; Yan et al., 2021). For example, in most previous studies, ρ was set as 1.225 (kg/m³), the gravitational acceleration g was set as 9.807 (m/s²), the drag coefficient of blade C_b was set as 0.0008, the power transfer efficiency e_t was set as 0.7.

4.2 | Energy consumption during horizontal flight

According to the integrated model proposed in Section 3 and the collected parameter values for each sample RWD, we estimated the energy consumption during horizontal flight. The estimated values of the energy consumption and the actual battery capacity were compared to evaluate the accuracy of the integrated model. First, the thrust was calculated as Equation (71).

Then, the thrust value was substituted into the implicit function shown as Equation (75), to obtain the induced velocity v_{is} .

$$v_{is} = T_s / (2\rho A_p \sqrt{((v_s + a_s t_s) \cos \gamma_s)^2 + ((v_s + a_s t_s) \sin \gamma_s + v_{is})^2}) \quad (75)$$

$(\gamma_s = \alpha)$

Finally, the induced energy consumption ($E_{in} = P_{in}t$), profile energy consumption ($E_{pr} = P_{pr}t$), parasite energy consumption ($E_{pa} = P_{pa}t$), electronics energy consumption ($E_e = P_e t$), and total energy consumption were calculated.

We introduced GAP0 to denote the percentage gap between the estimated energy consumption and the battery capacity, that is, $GAP0 = 100\% \times (E_{battery} - E_s) / E_{battery}$. Table 2 lists the results for the energy consumption of the sample RWDs during horizontal flight. For all the RWDs, the estimated energy consumption was lower than the battery capacity. The largest, smallest, and average GAP0 values were 6.05%, 0.02%, and 2.74%, respectively.

In the literature reviewed in Section 2, several types of RWD energy-consumption models were presented. We selected the following models (denoted as Model-1 to Model-4) from the literature to estimate the energy consumption of the sample RWDs during horizontal flight. Model-1 (e.g., D'Andrea, 2014) uses the lift-to-drag ratio as the key factor affecting the RWD flight efficiency, which is expressed by Equation (4). The lift-to-drag ratio was used only in Model-1, and its value was set as 3 by referring to the work of D'Andrea (2014). Model-2 (e.g., Dorling et al., 2016), expressed by Equation (6), assumes that the energy consumption for horizontal flight is approximately equal to the induced energy consumption for hovering. The horizontal airspeed of the RWD is ignored; thus, the energy consumption estimated by this model is biased. Model-3 (e.g., Sorbelli et al., 2020), which is expressed by Equations (10), (11), and (13), takes the induced power as the power consumption. Unlike Model-2, Model-3 considers the horizontal airspeed and the angle of attack. However, Model-3 ignores the effects of the profile power and parasite power. Model-3 may not estimate the power consumption accurately when the horizontal airspeed is high, as the parasite

TABLE 2 Results for the energy consumption during horizontal flight.

	RWD1	RWD2	RWD3	RWD4	RWD5	RWD6	RWD7	RWD8	RWD9	RWD10
T_s (N)	16.13	12.29	7.29	8.90	8.78	5.59	11.93	13.53	166.72	15.69
v_{is} (m/s)	4.69	3.42	3.11	2.33	3.73	3.54	3.61	3.94	6.64	5.69
E_{in} (J)	210,000	113,967	67,523	64,307	149,957	66,551	106,267	157,686	1,859,407	168,274
E_{in}/E_s (%)	73.32	42.86	44.27	31.50	57.58	45.96	45.76	49.70	76.41	69.78
E_{pr} (J)	74,421	137,349	69,053	120,610	83,464	65,111	113,779	136,534	557,564	59,190
E_{pr}/E_s (%)	25.98	51.65	45.27	59.08	32.05	44.97	49.00	43.04	22.90	24.55
E_{pa} (J)	1042	3139	529	924	1385	187	168	202	4766	985
E_{pa}/E_s (%)	0.36	1.18	0.35	0.45	0.53	0.13	0.07	0.06	0.20	0.41
E_e (J)	960	11,479	15,429	18,318	25,629	12,939	12,000	22,834	11,803	12,703
E_e/E_s (%)	0.34	4.32	10.12	8.97	9.84	8.94	5.17	7.20	0.49	5.27
E_s (J)	286,420	265,934	152,534	204,159	260,436	144,789	232,214	317,256	2,433,539	241,152
GAP0 (%)	0.30	5.84	2.82	4.35	6.05	0.50	5.14	1.20	1.17	0.02

TABLE 3 Comparison of rotary-wing drone energy-consumption models.

Model	Mechanism	Acceleration or deceleration considered?	Flight phases considered?	Average GAP0 (%)
Model-1	Use the lift-to-drag ratio	Not	Not	65.00
Model-2	Energy consumption for horizontal flight equals the induced energy consumption for hovering. Ignore the horizontal airspeed	Not	Not	27.73
Model-3	Take the induced power as the power consumption. Ignore the profile power and parasite power	Not	Not	48.50
Model-4	Consider several types of power consumption. The angle of attack is ignored	Not	Yes	5.64
Integrated model	Use the actuator disc theory to analyze the dynamic process. Consider the angle of attack	Yes	Yes	2.74

power increases sharply with an increase in the horizontal airspeed. Model-4 (e.g., Kirschstein, 2020), expressed by Equation (33), considers several types of power consumption. However, in the estimation of the induced power, Model-4 does not include the angle of attack during horizontal flight. Unlike Model-4, the integrated model proposed in Section 3 uses the acceleration and the angle of attack to determine the power; thus, it is more effective and can be applied to flight attitudes with higher angles of attack. Table 3 summarizes characteristics of different types of models.

Table A3 presents the results for the estimated energy consumption obtained using various models. For Model-1, the largest, smallest, and average GAP0 values were 81.18%, 49.33%, and 65.00%, respectively. For Model-2, the largest, smallest, and average GAP0 values were 49.04%, 5.36%, and 27.73%, respectively. For Model-3, the largest, smallest, and average GAP0 values were 65.16%, 33.83%, and 48.50%, respectively. For Model-4, the largest, smallest, and average GAP0 values were 10.31%, 2.09%, and 5.64%, respectively. A larger GAP0 corresponds to more unutilized battery

energy, along with a lower usage ratio of the battery capacity. In computational experiments involving several sample RWDs, the integrated model outperformed Model-1 to Model-4 in estimating the energy consumption during horizontal flight, from the perspective of the usage ratio of the battery capacity.

To further examine the varying behaviors of Model-1 to Model-4 and the integrated model proposed in Section 3, the GAP0 results were arranged in the form of a boxplot, as shown in Figure 5. For the integrated model, the GAP0 values were in the range of (0%, 7%), and this range was the narrowest among the models. The GAP0 range became wider in the order of Model-4, Model-2, Model-3, and Model-1.

4.3 | Energy consumption of one flight segment

The integrated model (i.e., Equation 74) was used to calculate the energy consumption under the condition that an RWD completed

one flight segment (including climbing, hovering, horizontal flight, and descending) and the battery energy was approximately exhausted. In addition to the parameters mentioned in Section 4.1, the values of several parameters, including the distance of acceleration or deceleration (d_a) and altitude in horizontal flight (h_s), were obtained by consulting companies. In the computational experiment, d_a was 30 m, and h_s was 120 m.

For the sample RWDs, the computational results of the energy consumption of one flight segment are presented in Table 4. From the left, the second and third columns present the energy consumed during climbing and its percentage of the total energy consumption, respectively. The fourth and fifth columns present the energy consumed during horizontal flight and its percentage of the total energy consumption, respectively. The sixth and seventh columns present the energy consumed during descending and its percentage of the total energy consumption, respectively. The eighth and ninth columns present the energy consumed during hovering and its

percentage of the total energy consumption, respectively. From the viewpoint of battery capacity usage, on average, 87.63% of the battery capacity was consumed in the horizontal flight phase, which indicated that it is feasible to use the estimated energy consumption for the horizontal flight phase as the whole energy consumption. However, the actual energy consumption certainly exceeds the energy consumption in the horizontal flight phase; thus, an empirical correction factor (an upscaling factor with an average value of 1.14 and a range of [1.07, 1.31]) can be included in the estimation of the whole energy consumption.

5 | EFFECTS OF TOTAL MASS AND HORIZONTAL AIRSPEED ON RWD ENDURANCE

Considering the general and practical purpose of RWDs is to deliver parcels, we examined the effects of the total mass and horizontal airspeed on the RWD endurance. In this study, RWD endurance refers to the flight-time when an RWD completes one flight segment and the battery energy is nearly exhausted.

(i) In the first computational experiment, the total mass adopted in Section 4 (denoted as the benchmark mass) was increased and decreased by 10%, 30%, and 50%, and the other parameters remained unchanged. Table A4 presents the endurance for 10 sample RWDs with different masses. Figure 6 shows the endurance with respect to the total masses in the form of a boxplot.

We introduced GAP1 to denote the percentage gap between the endurance under the condition of the benchmark mass (denoted as $EDRW_b$) and that under the condition of a varied mass (denoted as $EDRW_c$), that is, $GAP1 = 100\% \times (EDRW_c - EDRW_b) / EDRW_b$. GAP1 was positive when the total mass was smaller than the benchmark mass and negative when the total mass was larger than the benchmark mass, which indicated that a larger mass corresponded to shorter endurance. For the 10 sample RWDs, the endurance

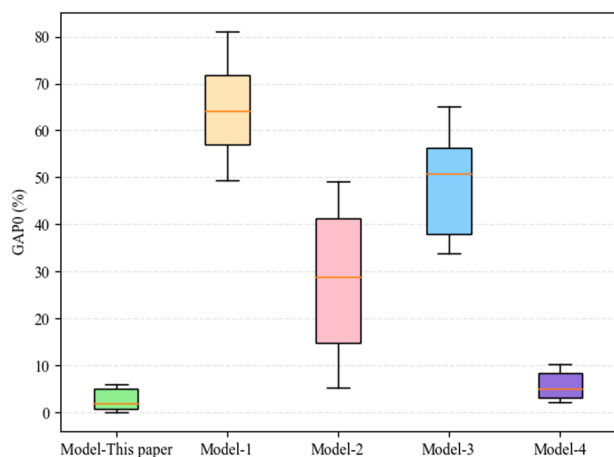


FIGURE 5 Performance of the energy-consumption models.

TABLE 4 Energy consumption for various flight phases.

	Climbing		Horizontal flight		Descending		Hovering	
	Value (J)	Percentage	Value (J)	Percentage	Value (J)	Percentage	Value (J)	Percentage
RWD1	11,441	3.98	252,817	88.00	19,025	6.62	3998	1.39
RWD2	11,524	4.08	246,510	87.28	20,121	7.12	4283	1.52
RWD3	4472	2.85	141,573	90.20	8970	5.71	1944	1.24
RWD4	5730	2.68	193,629	90.72	11,553	5.41	2532	1.19
RWD5	4612	1.66	259,522	93.62	10,772	3.89	2294	0.83
RWD6	4159	2.86	132,917	91.34	6929	4.76	1506	1.04
RWD7	11,580	4.73	215,486	88.03	14,559	5.95	3175	1.30
RWD8	7146	2.23	293,893	91.52	16,498	5.14	3583	1.12
RWD9	275,293	11.18	1,884,050	76.51	250,946	10.19	52,112	2.12
RWD10	21,389	8.87	190,675	79.05	24,110	10.00	5026	2.08

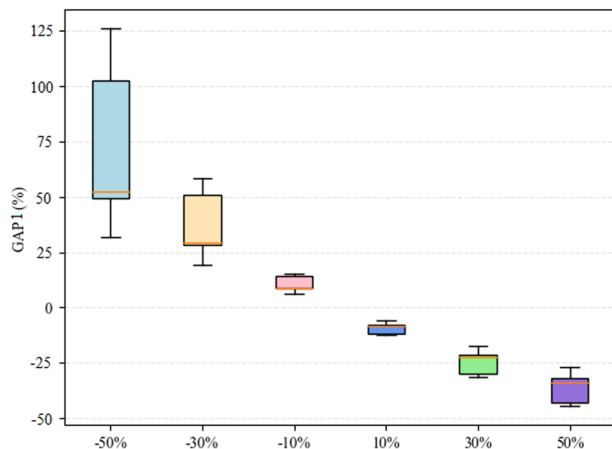


FIGURE 6 Boxplot of the rotary-wing drone endurance with respect to the total mass.

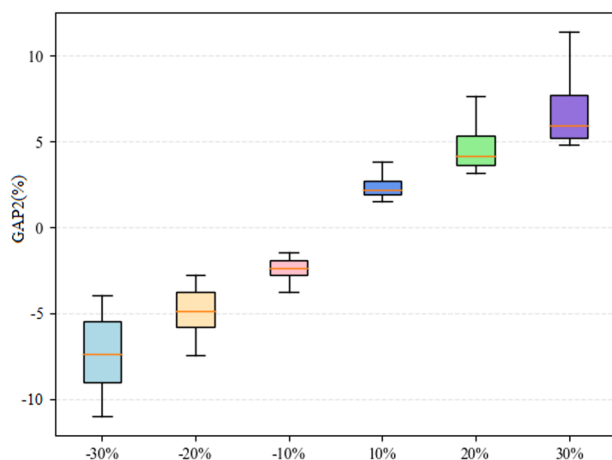


FIGURE 7 Boxplot of the rotary-wing drone endurance with respect to the horizontal airspeed.

generally increased by 50%–100% if the total mass was reduced by 50%, increased by 25%–50% if the total mass was reduced by 30%, decreased by 25%–50% if the total mass was increased by 50%, and varied within 15% if the total mass was changed by $\leq 10\%$.

(ii) In the second computational experiment, we examined the effect of the horizontal airspeed on the RWD endurance. The horizontal airspeed adopted in Section 4 (denoted as the benchmark horizontal airspeed) was increased and decreased by 10%, 20%, and 30%, and the other parameters remained unchanged. Table A5 presents the endurance for the 10 sample RWDs with different horizontal airspeeds. Figure 7 shows the endurance with respect to the horizontal airspeed in the form of a boxplot.

We introduced GAP2 to denote the percentage gap between the RWD endurance under the condition of the benchmark horizontal airspeed (denoted as $EDRS_b$) and that under the condition of a varied horizontal airspeed (denoted as $EDRS_c$), that is, $GAP2 = 100\% \times$

$(EDRS_c - EDRS_b)/EDRS_b$. GAP2 was positive when the horizontal airspeed was lower than the benchmark value and negative when the horizontal airspeed was higher than the benchmark value, which indicated that in the experimental range of the horizontal airspeed, a higher horizontal airspeed corresponded to longer endurance. For the 10 sample RWDs, the endurance generally decreased by 5%–10% if the horizontal airspeed was reduced by 30%, increased by 5%–8% if the horizontal airspeed was increased by 30%, decreased by 5% if the horizontal airspeed was reduced by 20%, and varied within 8% if the horizontal airspeed was changed within 10%.

(iii) In the third computational experiment, we comprehensively examined the effects of the horizontal airspeed and total mass on the RWD endurance. The benchmark mass was increased and decreased by 10%, 30%, and 50%, and the benchmark horizontal airspeed was increased and decreased by 10%, 20%, and 30%. Table A6 presents the endurance for the 10 sample RWDs with different masses and horizontal airspeeds. Figure 8 shows the endurance with respect to the mass and horizontal airspeed in the form of a boxplot.

For a certain horizontal airspeed, we introduced GAP3 to denote the percentage gap between the endurance under the condition of the benchmark mass (denoted as $EDRSW_b$) and that under the condition of a varied mass (denoted as $EDRSW_c$), that is, $GAP3 = 100\% \times (EDRSW_c - EDRSW_b)/EDRSW_b$. For each of the experimental horizontal airspeeds, GAP3 was positive when the total mass was smaller than the benchmark mass and negative when the total mass was larger than the benchmark mass. For the 10 sample RWDs, the endurance generally increased by 50%–125% if the mass was reduced by 50%, increased by 25%–50% if the mass was reduced by 30%, and varied within 15% if the mass was changed within 10%. Moreover, the total mass affected the endurance more significantly than the horizontal airspeed. For example, the RWD endurance generally varied within 15% if the horizontal airspeed was changed within 30%, whereas the RWD endurance generally varied by 15%–65% if the mass was changed within 30%.

6 | CONCLUSIONS

The RWD energy consumption is a critical factor that affects RWD route planning and delivery service quality. However, accurately estimating RWD energy consumption is very challenging. By analyzing the dynamic equilibrium of forces and power consumption of RWD in different flight stages, an RWD energy consumption model is established, which is a reproducible and versatile method to estimate the RWD energy consumption. From the methodological perspective, the integrated model can be referred to estimate the RWD flight time or distance, and load capacity to ensure the mission completion and to design the charging strategies. Considering the RWD energy consumption formula, designers can pursue optimizing the RWD structure and improving efficiency. Moreover, the integrated model can act as a supplementary and calibration way for measurement experiments. Computational experiments show that the integrated model is more effective than several models

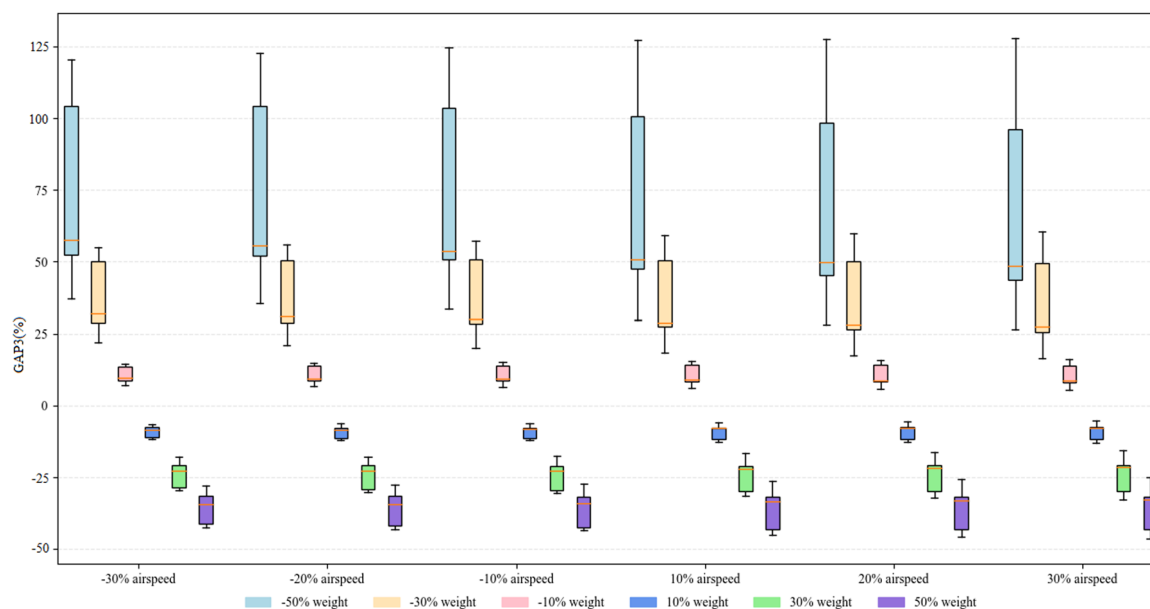


FIGURE 8 Boxplot of the rotary-wing drone endurance with respect to the total mass and horizontal airspeed.

proposed in the literature. Computational results indicated some interesting conclusions. For example, (i) although it is feasible to use the estimated energy consumption in the horizontal flight phase as the total energy consumption of an RWD flight segment, an empirical correction factor (an upscaling factor that can be obtained by computational experiments) should be included; (ii) a larger mass corresponds to shorter endurance; and (iii) the total mass affects the endurance more significantly than the horizontal airspeed.

To the best of our knowledge, the method expressed by Equation (74) is one of the most systematic ways for estimating the energy consumption of one flight segment. Although Equation (74) is too complex to draw directly conclusions on the energy-consumption variation with respect to some factors, and is computationally intensive to be directly involved in mathematical programming models for optimizing drone or truck-drone routes with consideration of the energy consumption, interesting future research can focus on numerical simulation based on Equation (74) and sample RWDs, so that suitable technical parameters are selected for RWD design, and suitable energy-consumption formulation is derived for mathematical programming models for RWD delivery systems. Besides, how to integrate the propulsion system into the integrated model remains an interesting research direction; physical experiments with various test conditions for validating the proposed model can be performed.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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APPENDIX A

TABLE A1 Parameters and variables.

Notation	Meaning
A_f	Projected frontal area
A_p	Propeller disc area
A_1	Cross-sectional area where airflow enters streamtube
A_2	Cross-sectional area where airflow is exhausted from bottom of streamtube
$a_c/a_d/a_s$	Acceleration during climbing/descending/forward horizontal flight
C_b/C_d	Drag coefficient of blade/RWD body
$C_{in}/C_{pr}/C_{pa}$	Coefficient of induced/profile/parasite drag
C_w	Downwash coefficient
D	Parasite drag
D_s	Drag force in horizontal flight
d_a	Distance of RWD acceleration or deceleration
d_s/d_v	Horizontal/Vertical flight distance
E	Total energy consumption
$E_{battery}$	Battery energy
$E_c/E_d/E_h$	Energy consumption during climbing/descending/ hovering
E_e	Energy consumption of electronics
$E_{in}/E_{pa}/E_{pr}$	Induced/parasite/profile energy consumption
E_s/E_v	Energy consumption in horizontal/vertical flight
E_h^t	Energy consumption per unit hovering time
$E_c^d/E_s^d/E_d^d$	Energy consumption per unit climbing/horizontal/ descending distance
e_o	Oswald efficiency (wingspan efficiency), with typical value between 0.70 and 0.85
e_t	Power transfer efficiency
G	Gravitational force
g	Gravitational acceleration
h_i	Flight height ($i = 1, 2, 3$)
h_s	Altitude in horizontal flight
k_i	Coefficient ($i = 1, 2, \dots, 11$)
L	Lift
m	Total mass
$m_b/m_d/m_p$	Mass of battery/RWD/payload
N_b	Total number of blades in single propeller
P	Total power consumption
$P_c/P_d/P_h/P_s$	Power consumption for climbing/descending/ hovering/horizontal flight

TABLE A1 (Continued)

Notation	Meaning
P_e	Power consumption of electronics
$P_{in}/P_{pa}/P_{pr}$	Induced/parasite/profile power
P_{lift}	Power consumption for lifting
R_r	Radius of signal rotor
r_d	Fuselage drag ratio
r_{ld}	Lift-to-drag ratio
r_w	Aspect ratio of wing
s_r	Rotor solidity
T	Thrust
T_c/T_{cv}	Thrust during climbing/vertical takeoff
T_d/T_{dv}	Thrust during descending/vertical descending
T_h/T_s	Thrust during hovering/horizontal flight
t	Total flight time of one flight segment
$t_c/t_d/t_h/t_s$	Climbing/descending/hovering/horizontal flight time
t_{max}	Maximum flight time measured in laboratory environment
w_b	Blade chord width
w_d	RWD width
v	Given airspeed
v_c/v_{cv}	Climbing airspeed/vertical-takeoff airspeed
v_d/v_{dv}	Descending airspeed/vertical-descending airspeed
v_{ic}/v_{icv}	Induced velocity during climbing/vertical takeoff
v_{id}/v_{idv}	Induced velocity during descending/vertical descending
v_{ih}/v_{is}	Induced velocity during hovering/horizontal flight
v_r	Rotor blade tip speed
v_s/v_v	Horizontal/vertical airspeed
v_0	Initial airspeed (when acceleration is considered in horizontal flight)
v_1	Velocity of airflow that enters streamtube
v_2	Velocity of airflow exhausted from bottom of streamtube
α	Tilt angle of the rotor disc
$\gamma_c/\gamma_d/\gamma_s$	Body pitch angle during climbing/descending/ horizontal flight
θ_c/θ_d	Flight trajectory angle in climbing/descending direction
ρ	Air density
ω_b	Angular speed of blade
κ	Upscaling factor (usually equal to 1.15)

TABLE A2 Parameter values of the sample rotary-wing drones.

Parameter	RWD1	RWD2	RWD3	RWD4	RWD5	RWD6	RWD7	RWD8	RWD9	RWD10
m (kg)	1.645	1.253	0.743	0.907	0.895	0.570	1.216	1.380	17	1.6
A_p (m ²)	0.1443	0.1017	0.1256	0.2123	0.0983	0.1050	0.2500	0.2500	0.9200	0.0900
v_r (m/s)	120	167	120	120	120	120	120	120	150	148
P_e (W)	0.4	5	6	6.894	6.5	4.44	5.6	8.88	8.1	7.8
v_c (m/s)	4	4	5	5	6	4	5	6	2	2.5
v_d (m/s)	2	2	2	2	2	2	2	2	2	2
v_s (m/s)	8.5	14	6.94	6.94	9	5	4	4	9	11.11
t_{max} (s)	1680	1607	1800	1860	2760	2040	1500	1800	1020	1140
$E_{battery}$ (J)	287,280	282,438	156,960	213,444	277,200	145,512	244,800	321,120	2,462,400	241,200

TABLE A3 Results for the energy consumption estimated by several models.

	Model-1		Model-2		Model-3		Model-4	
	E_s (J)	GAP0 (%)	E_s (J)	GAP0 (%)	E_s (J)	GAP0 (%)	E_s (J)	GAP0 (%)
RWD1	110,661	61.48	262,493	8.63	183,704	36.05	280,385	2.40
RWD2	143,130	49.33	309,600	25.79	110,990	60.70	253,330	10.31
RWD3	58,773	62.56	106,610	32.08	74,214	52.72	147,470	6.04
RWD4	72,994	65.80	116,060	45.63	74,358	65.16	165,320	8.49
RWD5	129,450	53.30	234,510	15.40	156,210	43.65	253,400	8.59
RWD6	40091	72.45	88,880	38.92	70,835	51.32	140,430	3.49
RWD7	46072	81.18	124,760	49.04	104,430	57.34	224,820	8.16
RWD8	69235	78.44	186,420	41.95	159,980	50.18	308,380	3.97
RWD9	740,603	69.92	2,106,107	14.47	1,629,300	33.83	2,392,064	2.86
RWD10	107,339	55.50	228,274	5.36	159,157	34.02	236,160	2.09

TABLE A4 Rotary-wing drone endurance (unit: s) for different masses.

	Total mass						
	-50%	-30%	-10%	benchmark	+10%	+30%	+50%
RWD1	3533.62	2537.20	1871.55	1628.09	1428.21	1124.77	909.97
RWD2	2445.01	2099.96	1772.47	1623.77	1486.87	1248.78	1054.53
RWD3	2709.52	2326.16	1971.32	1812.74	1667.88	1417.43	1214.38
RWD4	2508.55	2271.92	2025.88	1906.36	1791.53	1579.60	1393.64
RWD5	5039.21	4040.44	3221.97	2885.30	2592.10	2115.88	1754.72
RWD6	3032.08	2590.52	2197.53	2025.12	1868.74	1600.13	1381.96
RWD7	2315.41	1984.19	1692.51	1564.68	1448.56	1248.34	1084.64
RWD8	2774.92	2337.35	1965.94	1806.81	1664.04	1421.65	1226.82
RWD9	2163.21	1512.97	1102.98	956.70	837.75	658.73	532.81
RWD10	2221.27	1620.99	1204.59	1049.37	920.88	724.22	584.08

	Horizontal airspeeds						
	-30%	-20%	-10%	benchmark	+10%	+20%	+30%
RWD1	1460.41	1511.93	1568.28	1628.09	1689.89	1752.23	1813.80
RWD2	1473.81	1529.00	1579.48	1623.77	1661.19	1691.62	1715.19
RWD3	1669.90	1716.53	1764.71	1812.74	1859.20	1903.01	1943.45
RWD4	1777.34	1822.46	1865.97	1906.36	1942.73	1974.64	2002.01
RWD5	2568.81	2671.27	2778.02	2885.30	2989.87	3089.24	3181.62
RWD6	1914.70	1948.69	1985.75	2025.12	2065.93	2107.33	2148.49
RWD7	1502.07	1520.98	1541.94	1564.68	1588.87	1614.13	1640.07
RWD8	1735.50	1756.88	1780.72	1806.81	1834.84	1864.46	1895.28
RWD9	904.99	921.16	938.53	956.70	975.22	993.64	1011.53
RWD10	963.65	991.07	1019.96	1049.37	1078.41	1106.27	1132.31

TABLE A5 Rotary-wing drone endurance (unit: s) for different horizontal airspeeds.

TABLE A6 Rotary-wing drone endurance (unit: s) for different horizontal airspeeds and total masses.

	Airspeed	Mass					
		-50%	-30%	-10%	+10%	+30%	+50%
RWD1	-30%	3183.02	2256.88	1671.10	1288.03	1026.18	839.71
	-20%	3305.41	2347.81	1733.51	1330.68	1055.79	860.66
	-10%	3423.68	2442.37	1800.96	1377.73	1088.82	884.17
	+10%	3632.64	2629.41	1943.37	1481.01	1163.02	937.72
	+20%	3719.43	2716.77	2014.64	1534.98	1202.88	966.99
	+30%	3793.51	2797.72	2083.87	1589.02	1243.62	997.32
RWD2	-30%	2362.01	1966.55	1621.83	1341.65	1119.99	945.58
	-20%	2400.80	2020.96	1678.86	1393.82	1164.48	982.13
	-10%	2428.05	2065.43	1729.45	1442.78	1207.97	1018.93
	+10%	2452.86	2125.12	1807.58	1525.18	1285.78	1087.85
	+20%	2452.64	2141.61	1834.92	1557.35	1318.32	1118.10
	+30%	2445.19	2150.20	1854.86	1583.32	1346.07	1144.85
RWD3	-30%	2587.41	2175.00	1821.53	1534.01	1303.97	1120.01
	-20%	2634.13	2228.28	1871.46	1576.97	1339.53	1148.98
	-10%	2675.02	2279.24	1922.01	1622.08	1377.80	1180.67
	+10%	2737.71	2368.08	2017.97	1713.01	1458.26	1249.34
	+20%	2759.96	2404.62	2061.02	1756.37	1498.40	1284.76
	+30%	2776.75	2435.74	2099.90	1797.16	1537.33	1319.93
RWD4	-30%	2439.91	2165.27	1900.13	1662.41	1456.90	1282.19
	-20%	2468.64	2205.96	1945.22	1706.60	1497.43	1318.02
	-10%	2491.38	2241.69	1987.58	1750.17	1538.83	1355.59
	+10%	2520.72	2296.66	2059.49	1829.61	1618.54	1431.03
	+20%	2528.44	2316.23	2088.21	1863.78	1654.79	1466.86
	+30%	2532.18	2331.01	2112.14	1893.79	1687.82	1500.47

TABLE A6 (Continued)

	Airspeed	Mass					
		−50%	−30%	−10%	+10%	+30%	+50%
RWD5	−30%	4685.02	3651.95	2875.09	2306.89	1888.64	1575.27
	−20%	4820.19	3787.95	2989.77	2397.65	1959.12	1629.96
	−10%	4938.89	3918.97	3106.84	2493.76	2035.55	1690.21
	+10%	5121.09	4149.64	3331.74	2689.76	2197.99	1822.04
	+20%	5185.32	4245.21	3433.75	2784.30	2279.87	1890.66
	+30%	5232.99	4326.73	3526.46	2873.88	2359.82	1959.19
RWD6	−30%	2908.88	2459.40	2078.38	1767.49	1516.54	1313.69
	−20%	2951.29	2501.78	2115.51	1798.36	1541.69	1334.06
	−10%	2992.84	2545.96	2155.55	1832.31	1569.67	1356.89
	+10%	3068.03	2634.19	2240.42	1906.95	1632.64	1409.03
	+20%	3100.21	2675.94	2283.23	1946.22	1666.69	1437.75
	+30%	3128.44	2715.03	2325.11	1985.80	1701.75	1467.77
RWD7	−30%	2238.49	1907.09	1624.41	1391.46	1201.48	1046.46
	−20%	2263.68	1931.17	1645.15	1408.60	1215.43	1057.77
	−10%	2289.61	1957.07	1667.97	1427.71	1231.09	1070.51
	+10%	2340.36	2011.87	1718.35	1470.93	1267.05	1100.06
	+20%	2363.89	2039.50	1745.03	1494.50	1287.03	1116.67
	+30%	2385.65	2066.53	1772.08	1518.96	1308.08	1134.35
RWD8	−30%	2677.59	2244.91	1887.27	1599.79	1369.92	1185.25
	−20%	2708.86	2273.40	1911.01	1618.96	1385.24	1197.51
	−10%	2741.63	2304.42	1937.34	1640.43	1402.52	1211.39
	+10%	2807.81	2371.52	1996.42	1689.58	1442.53	1243.76
	+20%	2839.52	2406.23	2028.30	1716.78	1465.02	1262.13
	+30%	2869.48	2440.81	2061.11	1745.33	1488.91	1281.80
RWD9	−30%	1994.34	1401.44	1035.55	798.41	636.78	521.71
	−20%	2050.75	1436.91	1056.66	810.77	643.84	525.50
	−10%	2107.77	1474.46	1079.31	823.98	651.22	529.25
	+10%	2215.38	1551.28	1127.07	851.74	666.16	536.05
	+20%	2263.08	1588.39	1150.96	865.60	673.28	538.79
	+30%	2305.61	1623.46	1174.10	879.01	679.89	540.88
RWD10	−30%	2037.43	1470.61	1099.43	851.72	680.32	557.34
	−20%	2103.97	1521.17	1133.44	873.70	694.26	565.95
	−10%	2165.91	1571.96	1168.90	896.99	709.03	574.96
	+10%	2269.08	1666.77	1239.37	944.66	739.38	593.03
	+20%	2309.09	1708.30	1272.31	967.66	754.09	601.53
	+30%	2341.40	1745.00	1302.72	989.33	767.95	609.31